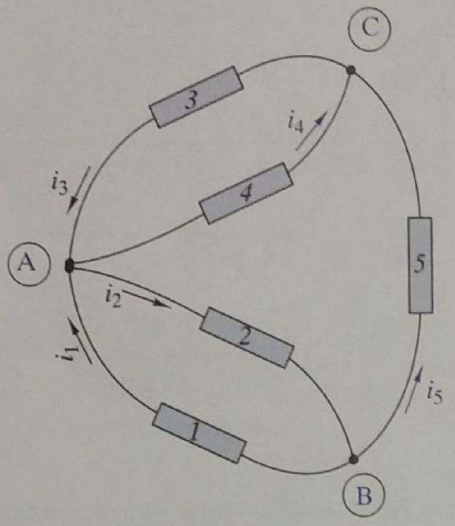

FIGURE P2–22

2–23 **(a)** Use the passive sign convention to assign voltage variables consistent with the currents in Figure P2-22. Write three KVL connection equations using these voltage variables.
(b) If $v_4 = 0$ V, what can be said about the voltages across all the other elements?

2–24 The KCL equations for a three-node circuit are:

Node A $\quad -i_1 + i_2 - i_4 = 0$
Node B $\quad -i_1 - i_3 + i_5 = 0$
Node C $\quad i_1 + i_3 + i_4 - i_5 = 0$

Draw the circuit diagram and indicate the reference directions for the element currents.

OBJECTIVE 2–3 COMBINED CONSTRAINTS (SECT. 2–3)

Given a linear resistance circuit, use the element constraints and connection constraints to find selected signal variables. See Examples 2–8, 2–9, 2–10, 2–11, and 2–12 and Exercise 2–9, 2–10, 2–11, 2–12, and 2–13.

2–25 Find v_x and i_x in Figure P2–25.

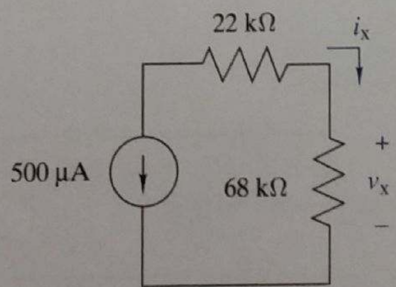

FIGURE P2–25

2–26 Find v_x and i_x in Figure P2–26.

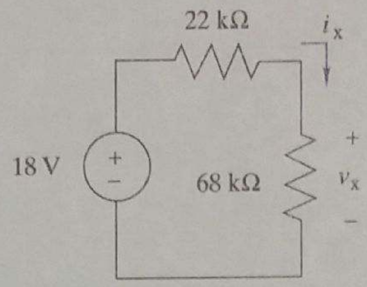

FIGURE P2–26

2–27 Find v_x and i_x in Figure P2–27. Compare the results of your answers with those in problem 2–26. What effect did adding the 33-kΩ resistor have on the overall circuit? Why isn't i_y zero?

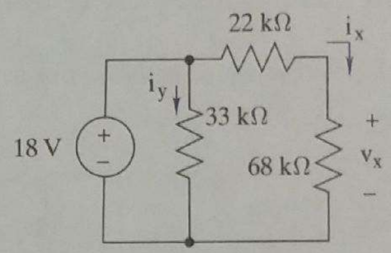

FIGURE P2–27

2–28 A modeler wants to light his model building using miniature grain-of-wheat light bulbs connected in parallel as shown in Figure P2–28. He uses two 1.5-V "C-cells" to power his lights. He wants to use as many lights as possible, but wants to limit his current drain to 500 μA to preserve the batteries. If each light has a resistance of 36 kΩ, how many lights can he install and still be under his current limit?

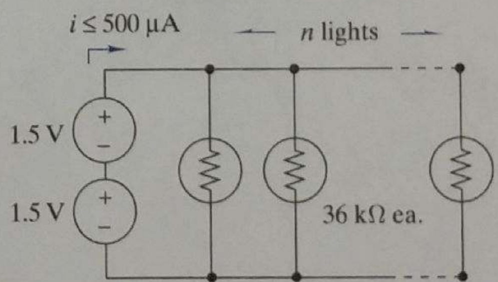

FIGURE P2–28

2–29 Find v_x and i_x in Figure P2–29.

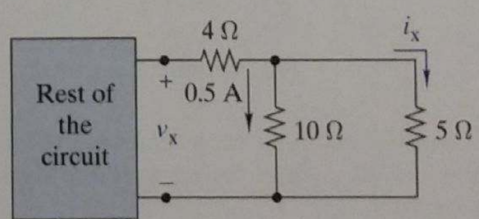

FIGURE P2–29

2–30 In Figure P2–30:

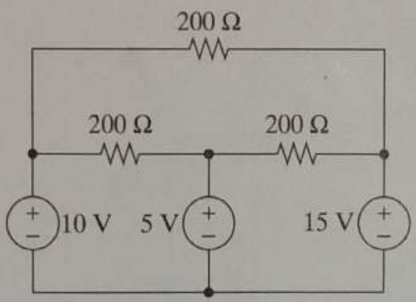

FIGURE P2–30

(a) Assign a voltage and current variable to every element.
(b) Use KVL to find the voltage across each resistor.
(c) Use Ohm's law to find the current through each resistor.
(d) Use KCL to find the current through each voltage source.

2–31 Find the power provided by the source in Figure P2–31.

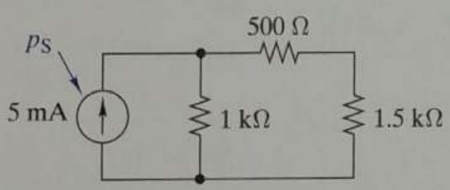

FIGURE P2–31

2–32 Figure P2–32 shows a subcircuit connected to the rest of the circuit at four points.

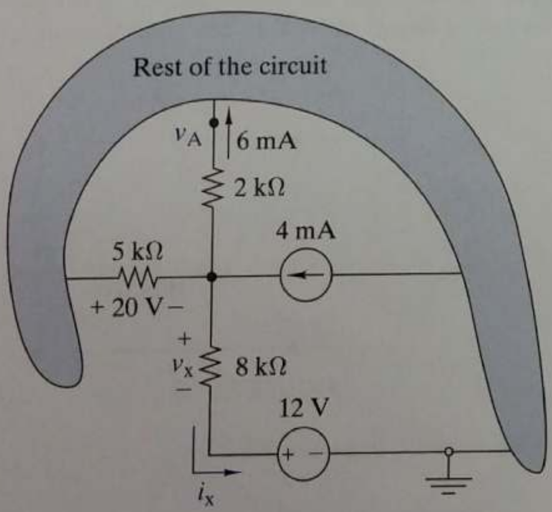

FIGURE P2–32

(a) Use element and connection constraints to find v_x and i_x.
(b) Show that the sum of the currents into the rest of the circuit is zero.
(c) Find the voltage v_A with respect to the ground in the circuit.

2–33 In Figure P2–33 i_x = 0.5 mA. Find the value of R.

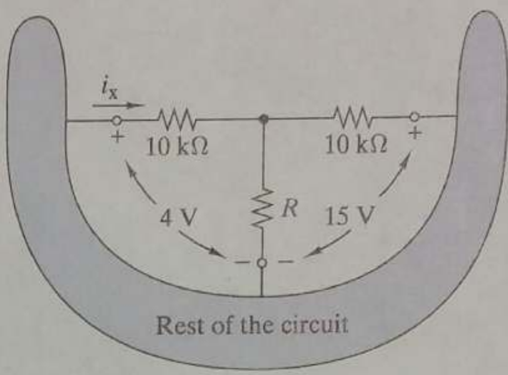

FIGURE P2–33

2–34 Figure P2–34 shows a resistor with one terminal connected to ground and the other connected to an arrow. The arrow symbol is used to indicate a connection to one terminal of a voltage source whose other terminal is connected to ground. The label next to the arrow indicates the source voltage at the ungrounded terminal. Find the voltage across, current through, and power dissipated in the resistor.

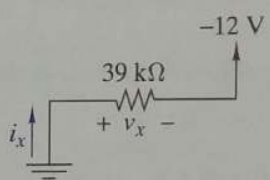

FIGURE P2–34

OBJECTIVE 2–4 EQUIVALENT CIRCUITS (SECT. 2–4)

(a) Given a circuit consisting of linear resistors, find the equivalent resistance between a specified pair of terminals.
(b) Given a circuit consisting of a source-resistor combination, find an equivalent source-resistor circuit.

See Example 2–13, 2–14, and 2–15 and Exercises 2–14, 2–15, 2–16, 2–17, and 2–18.

2–35 Find the equivalent resistance R_{EQ} in Figure P2–35.

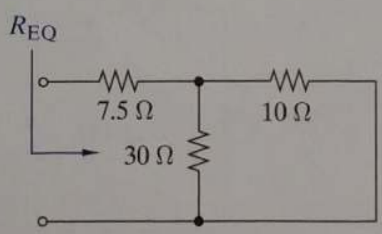

FIGURE P2–35

2–36 Find the equivalent resistance R_{EQ} in Figure P2–36.

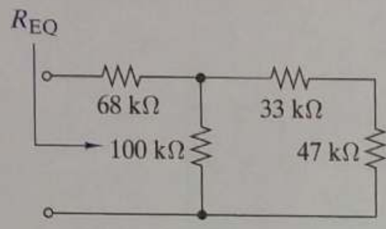

FIGURE P2–36

2–37 Find the equivalent resistance if R_{EQ} in Figure P2–37.

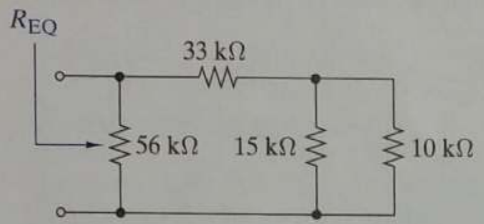

FIGURE P2–37

2–38 Equivalent resistance is defined at a particular pair of terminals. In the following figure the same circuit is looked at from two different terminal pairs. Find the equivalent resistances R_{EQ1} and R_{EQ2} in Figure P2–38. Note that in calculating R_{EQ2} the 33-kΩ resistor is connected to an open circuit and therefore doesn't affect the calculation.

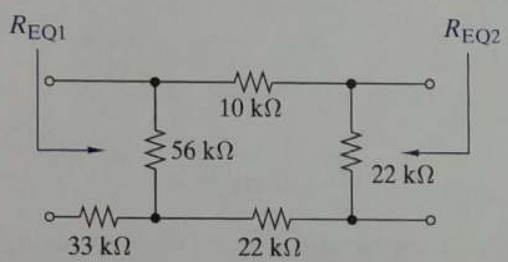

FIGURE P2–38

2–39 Find R_{EQ} in Figure P2–39 when the switch is open. Repeat when the switch is closed.

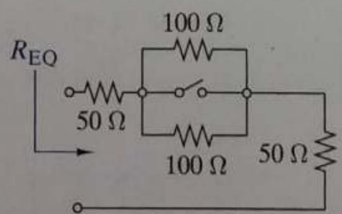

FIGURE P2–39

2–40 Find R_{EQ} between nodes A and B for each of the circuits in Figure P2–40. What conclusion can you draw about resistors of the same value connected in parallel?

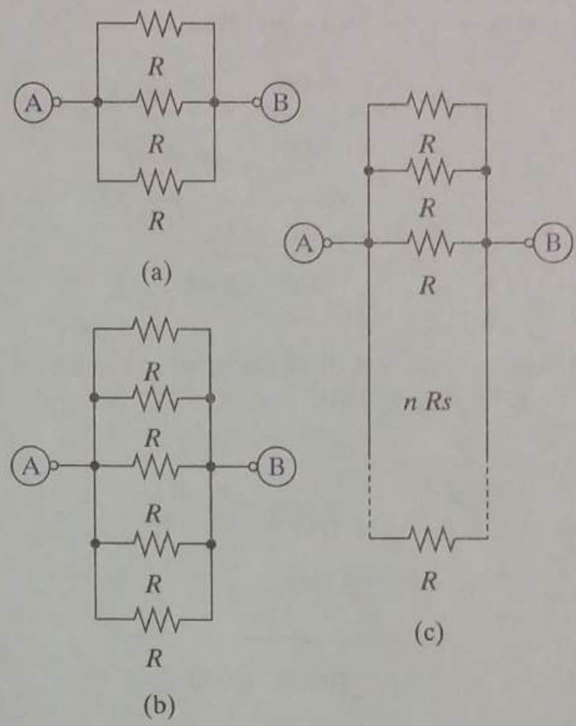

FIGURE P2–40

2–41 Show how the circuit in Figure P2–41 could be connected to achieve a resistance of 100 Ω, 200 Ω, 150 Ω, 50 Ω, 25 Ω, 33.3 Ω, and 133.3 Ω.

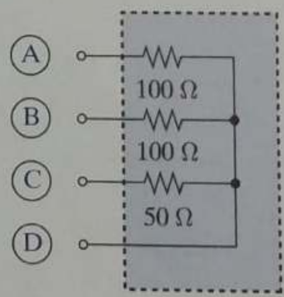

FIGURE P2–41

2–42 In Figure P2–42 find the equivalent resistance between terminals A-B, A-C, A-D, B-C, B-D, and C-D.

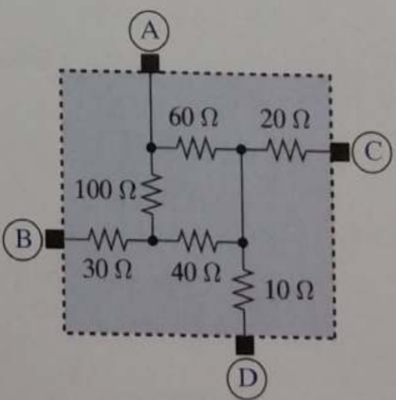

FIGURE P2–42

2–53 Find the equivalent resistance between terminals A and B in Figure P2–53.

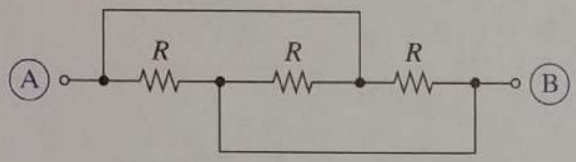

FIGURE P2–53

OBJECTIVE 2–5 VOLTAGE AND CURRENT DIVISION (SECT. 2–5)

(a) Given a linear resistive circuit with elements connected in series or parallel, use voltage or current division to find specified voltages or currents.
(b) Design a voltage or current divider that delivers specified outputs signals.

See Examples 2–16, 2–17, 2–18, 2–19, 2–20, 2–21, 2–22, and 2–23 and Exercises 2–19, 2–20, 2–21, 2–22, 2–23, 2–24, and 2–25.

2–54 Use voltage division in Figure P2–54 to find v_x.

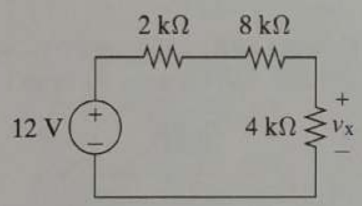

FIGURE P2–54

2–55 Use voltage division in Figure P2–55 to obtain an expression for v_L in terms of R, R_L, and v_S.

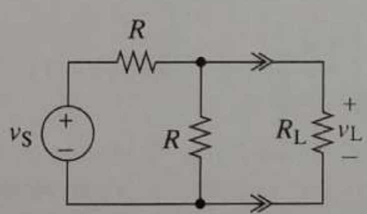

FIGURE P2–55

2–56 Use current division in Figure P2–56 to find i_x and v_x.

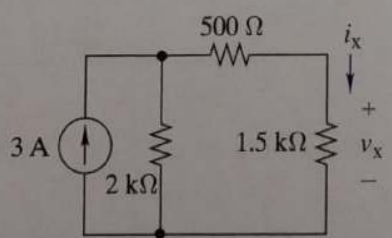

FIGURE P2–56

2–57 Use current division in Figure P2–57 to find an expression for v_L in terms of R, R_L and i_S.

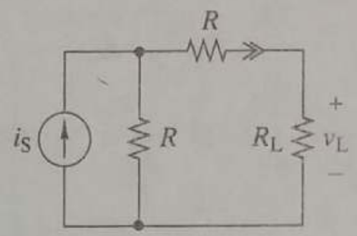

FIGURE P2–57

2–58 Find i_x, i_y, and i_z in Figure P2–58.

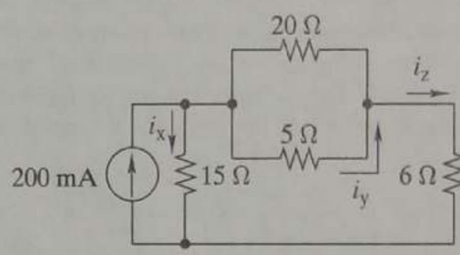

FIGURE P2–58

2–59 Find v_O in the circuit of Figure P2–59.

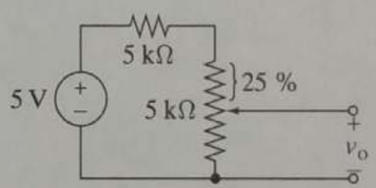

FIGURE P2–59

2–60 **A** The 1-kΩ load in Figure P2–60 needs 5 V across it to operate correctly. Where should the wiper on the potentiometer be set (R_x) to obtain the desired output voltage?

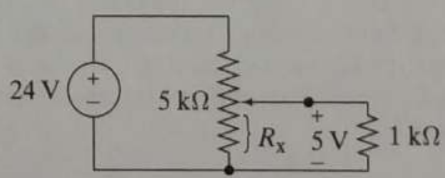

FIGURE P2–60

2–61 Find the range of values of v_O in Figure P2–61.

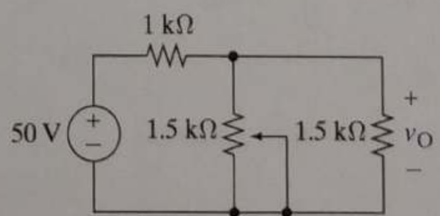

FIGURE P2–61

2–69 Use circuit reduction to find v_x, i_x, and p_x in Figure P2–69. Repeat using OrCAD.

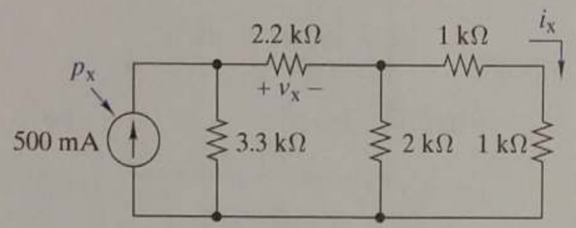

FIGURE P2–69

2–70 Use circuit reduction to find v_x and i_x in Figure P2–70.

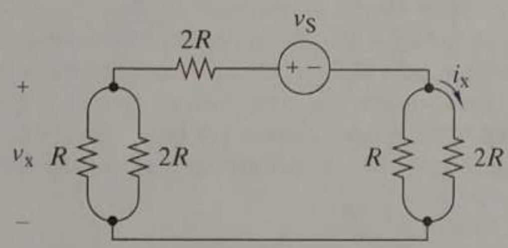

FIGURE P2–70

2–71 Use circuit reduction to find v_x, i_x, and p_x in Figure P2–71.

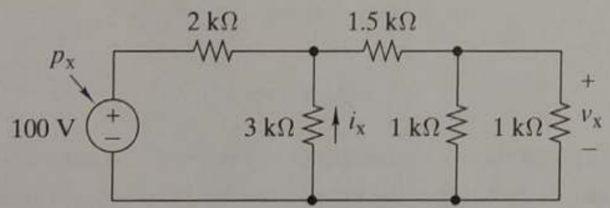

FIGURE P2–71

2–72 Use circuit reduction to find v_x and i_x in Figure P2–72.

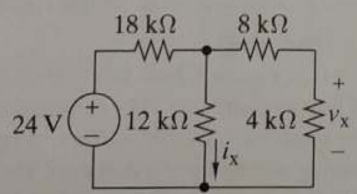

FIGURE P2–72

2–73 Use source transformation to find i_x in Figure P2–73.

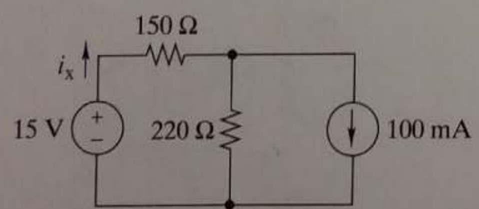

FIGURE P2–73

2–74 Select a value for R_x so that $i_x = 0$ A in Figure P2–74.

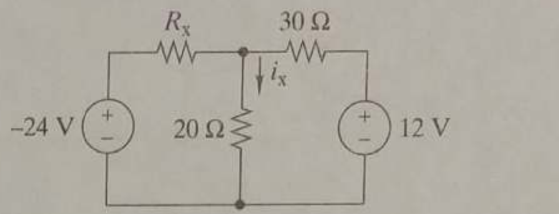

FIGURE P2–74

2–75 Use source transformations in Figure P2–75 to relate v_O to v_1, v_2, and v_3.

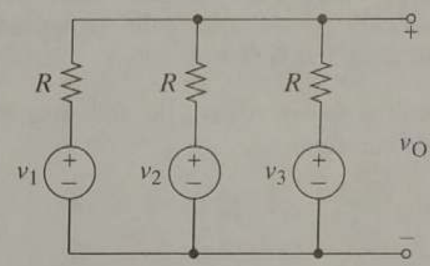

FIGURE P2–75

2–76 The current through R_L in Figure P2–76 is 100 mA. Use source transformations to find R_L. Validate your answer using OrCAD.

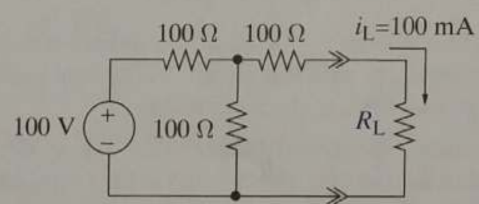

FIGURE P2–76

2–77 Select R_x so that 50 V are across it in Figure P2–77.

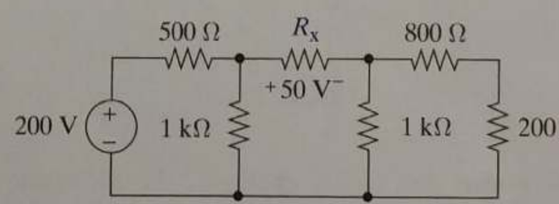

FIGURE P2–77

2–78 The box in the circuit in Figure P2–78 is a resistor whose value can be anywhere between 8 kΩ and 80 kΩ. Use circuit reduction to find the range of values of v_x.